Anxiety Disorders

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CPhA strives to provide guidance for health-care professionals on the appropriate care of all individuals. When the terms "women" and "men" are used in this chapter, they are used to maintain accuracy with what is reported in the clinical literature. We are aware these terms may not reflect the identity of the individual patient presenting and that they are not inclusive of all patients. Other resources exist that may be helpful, such as the World Professional Association for Transgender Health's (WPATH) Standard of Care.

Introduction

Anxiety disorders are among the most common psychiatric illnesses, with a lifetime prevalence estimated at 31%.[1] According to a 2014 Canadian survey on the sociodemographic characteristics associated with anxiety and mood disorders, about 24% of Canadians will report having suffered from an anxiety disorder in their lifetime.[2] Despite this significant prevalence, anxiety disorders are frequently underdiagnosed[3] and remain untreated in about 40% of diagnosed patients.[4][5]

Women are more likely than men to suffer from anxiety disorders. Age of onset varies: panic disorder and generalized anxiety disorder generally appear between 25 and 50 years of age, while separation anxiety disorder and phobias appear earlier, usually between 7 and 14 years of age.[1]

The current conceptualization of the etiology of anxiety disorders includes an interaction of psychosocial factors and a genetic vulnerability, which manifests in neurobiological and neuropsychological dysfunctions.[6] Among psychosocial risk factors, a family or personal history of anxiety or mood disorders or a personal history of trauma constitute significant predictors for the development of an anxiety disorder.[1] Isolation, a low level of education, parental deficiency or overprotection, and certain chronic physical illnesses such as cardiovascular disease or diabetes are all factors precipitating the occurrence of anxiety disorders.[1]

Anxiety disorders tend to run a chronic course. A study has shown that 58% of patients with generalized anxiety disorder and 31% of patients with panic disorder still have the diagnosis after 30 years.[7] For many of them, these disorders are associated with a high level of distress and functional impairment, which can be aggravated by significant physical or psychological comorbidity, particularly with depression and substance abuse, as well as hypertension, other cardiovascular diseases and gastrointestinal disorders. In addition, anxiety disorders are associated with significant social costs due to their adverse social and professional outcomes such as absenteeism and reduced performance at school or work.[8]

Table 1 presents anxiety disorders as defined in the *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision* (DSM-5-TR) as well as their specific characteristics.[9] A few reminders regarding the DSM-5-TR:

- Panic disorder and agoraphobia are no longer linked in a single diagnosis. These disorders have separate criteria, and patients who meet all of these criteria are now given 2 different diagnoses.
- Obsessive-compulsive disorder (OCD) and post-traumatic stress disorder (PTSD) are reclassified as separate disorders: obsessive-compulsive and related disorders, and trauma- and stressor-related disorders, respectively. Management of OCD and PTSD are discussed in Obsessive-Compulsive Disorder and .
- Two disorders, separation anxiety disorder and selective mutism, are included in anxiety disorders. Both disorders were previously classified in "disorders usually first diagnosed in infancy, childhood, or adolescence."

Goals of Therapy

- Eliminate or reduce symptomatic anxiety
- Eliminate or reduce anxiety-based disability
- Facilitate complete remission of symptoms and functional recovery
- Prevent recurrences
- Treat comorbid conditions

Investigations

- Thorough history with attention to:
  - nature and onset of symptoms
  - nature and extent of disability
  - presence of comorbid medical or psychiatric conditions
  - medication history and lifestyle habits (tobacco, alcohol, drugs, sources of caffeine)

Note: When a comorbid mood disorder is present, it is considered the main disorder, but both disorders must be treated simultaneously.

- Complete clinical evaluation to obtain an accurate diagnosis (see Table 1 and Table 2 as well as DSM-5-TR for diagnostic criteria[9])
- Validated clinical scales can be useful for improving symptom screening and detection, as well as for providing accurate and objective information on which to base clinical decisions (see Table 2). Several scales are available for assessing anxiety disorders and monitoring treatment response.[10] Examples include the Panic Disorder

Severity Scale (PDSS) for panic disorder, the Hamilton Anxiety Rating Scale (HAM-A) for generalized anxiety disorder and the Liebowitz Social Anxiety Scale (LSAS) for social anxiety disorder. These scales, however, are used only sporadically by clinicians in practice. The use of self-reported normalized scales may be more interesting because they are short, self-administered questionnaires that can quickly be completed by the patient to help monitor the progression of the anxiety disorder and evaluate the effectiveness of treatment. Some examples of self-reported questionnaires:
  - Anxiety in general: Hospital Anxiety and Depression Scale (HADS)
  - Panic disorder: Panic and Agoraphobia Scale (PAS)
  - Generalized anxiety disorder: Generalized Anxiety Disorder 7-item Scale (GAD-7)
  - Social anxiety disorder: Liebowitz Social Anxiety Scale (LSAS)
- Physical examination to rule out endocrine or cardiac disorders and to identify signs of substance use
- Laboratory tests: none routinely indicated; thyroid function if suspected

Note: Treat physical disorders of recent onset before making a definitive diagnosis of anxiety disorder.

***Table 1:** DSM-5–TR Classification of Anxiety Disorders*[9]

Type of Anxiety Disorder	Clinical Features
Separation anxiety disorder	Childhood onset of fear of separation from attachment figures (parents, siblings) that is excessive for the developmental stage.
Selective mutism	Childhood onset of failure to speak in school or other social situations when the individual does speak in other settings. May impede academic progression.
Specific phobia	Severe anxiety triggered by a specific feared object or situation (e.g., spiders, flying, heights) often leading to avoidance behaviour.
Social anxiety disorder (social phobia)	Intense anxiety provoked by social or performance situations in which embarrassment might occur; often leads to avoidance behaviour.
Panic disorder	Recurrent unexpected abrupt panic attacks with persistent anxiety concerning recurrence.
Agoraphobia	Marked fear or anxiety of 2 or more situations: public transportation, open spaces, closed spaces, crowds, being outside of home alone. Leads to avoidance of these situations.
Generalized anxiety disorder	Excessive worry and anxiety about a number of events or activities on more days than not over a period of ≥6 months.
Anxiety disorder due to another medical condition	Anxiety or panic attacks directly caused by a medical condition, e.g., thyroid dysfunction, hypoglycemia, heart failure, arrhythmia, COPD, vitamin B_{12} deficiency, encephalitis.
Substance/medication-induced anxiety disorder	Anxiety or panic attacks directly caused by use or discontinuation of a substance (e.g., alcohol, amphetamines, anticholinergics, caffeine, cannabis, cocaine, corticosteroids, hallucinogens) capable of producing the symptoms of anxiety.
Other specified anxiety disorder	Symptoms of anxiety disorders not meeting full diagnostic criteria, e.g., limited-symptom panic attacks, generalized anxiety occurring on fewer days than not.
Unspecified anxiety disorder	Distressing anxiety symptoms that fail to meet diagnostic criteria for specific anxiety disorders.

***Table 2:** Interview Questions for the Screening of Anxiety Symptoms*[1]

During the past 2 weeks, how much have you been bothered by the following problems?

- Feeling nervous, anxious, frightened, worried or on edge
- Feeling panic or being frightened
- Avoiding situations that make you anxious

Therapeutic Choices

A relatively mild state of anxiety can often occur in response to life circumstances. Incidentally, many patients manage their anxiety without psychotherapy or medication. Support, problem-solving and relaxation techniques or an approach based on mindfulness can sometimes be enough;[11] however, the persistence of stress factors or the accumulation of difficult life situations can lead to a more serious anxiety disorder sometimes requiring psychotherapy or the addition of pharmacotherapy. For patients suffering from an anxiety disorder, the therapeutic approach should include psychoeducation dealing specifically with the illness, its treatment, and its aggravating factors as well as the warning signs and symptoms of a relapse. Some internet-based educational resources can be helpful, such as Relief or Anxiety Canada.

Anxiety disorders can be effectively treated with psychotherapy and/or medication.

The treatment plan should be chosen only after careful consideration of individual factors such as patient preference, previous treatment attempts, severity of the anxiety disorder, presence of comorbidities such as major depressive episode or personality disorder, or presence of a substance use disorder or suicidal risk. Accessibility of different treatment plans and associated costs should also be considered.[6]

See Figure 1 for an illustration of the management of anxiety disorders.

Nonpharmacologic Choices

- **Psychotherapy** plays an important role in the management of anxiety disorders, with several meta-analyses having shown the effectiveness of **cognitive behavioural therapy (CBT)**,[12][13] **exposure therapy**[12][13] and **mindfulness-based therapy**.[14]
- Specific CBT tailored to the primary diagnosis is often the preferred psychotherapy. However, access to this type of therapy is sometimes limited in some areas. If the patient does not have access to traditional face-to-face CBT, internet-delivered CBT (ICBT) may be a reasonable alternative. A Cochrane meta-analysis found that ICBT resulted in a clinically significant improvement of anxiety compared with a waiting list, weekly virtual contact through online questionnaires or forums, or discussion groups.[15] In addition, no significant difference was found between face-to-face CBT, therapist-supported ICBT and unguided ICBT.[15] Nevertheless, further, more comprehensive studies are necessary to confirm these observations.[15] A comprehensive review of ICBT resulted in a list of available resources in Canada.[16]
- Stress-reduction techniques, which include relaxation and time-management training, are often helpful at the beginning of treatment.[17]
- Aerobic exercise several times per week can be recommended and may help to reduce some anxiety symptoms, although CBT remains significantly more effective.[18]
- Caffeine or other stimulant use should be reduced and monitored.
- Alcohol use should be minimal; more importantly, it should not be used to reduce anxiety. Alcohol induces sleep, but acts negatively on its quality and duration, especially by reducing deep sleep. Stopping or abruptly reducing excessive or chronic alcohol consumption may cause increased anxiety. The use of drugs such as cannabis, cocaine, hallucinogens and anabolic steroids may induce anxiety and should therefore be discontinued; refer the patient to an addiction treatment centre if necessary.
- It is critical to educate the patient about the importance of a balanced lifestyle and healthy sleep hygiene (see Insomnia).

Pharmacologic Choices

Pharmacotherapy is the most common intervention in the treatment of anxiety disorders when symptoms and disability are moderate to severe. A list of drugs used for the management of anxiety disorders is provided in Table 6.

Several antidepressants have demonstrated efficacy in the treatment of anxiety disorders. **SSRIs** and **SNRIs** are generally preferred as initial therapy due to their tolerability and safety profile compared to tricyclic antidepressants (**TCAs**) and **MAOIs**.[1][6]

The target antidepressant dose for anxiety disorders is similar to that used in major depressive disorder (see Depression). The antidepressant is initially introduced at a low dose to ensure tolerance, and then titrated every week or 2 until the usual dose for depression has been reached. Once the target dose is reached, some of the symptoms may improve after 2–8 additional weeks of treatment. However, optimal clinical response can take up to 8–12 weeks or more, which is slower than in the treatment of a major depressive episode. Patients must be informed that the adverse effects of medication often occur upon treatment initiation, while the beneficial effects on anxiety are only experienced later. Communicating this information to the patient is essential to promote adherence and improve medication efficacy. If there is no clinical response (response being defined as a 50% or greater improvement in score on a validated scale), switching to another antidepressant should be attempted before augmenting with a second agent since, despite the paucity of studies on this subject, clinical experience shows that patients can respond to an antidepressant from another class.[6]

Most patients suffering from anxiety disorders must continue pharmacotherapy for at least 12–24 months in order to achieve functional remission and prevent relapses. Moreover, a meta-analysis of relapse prevention trials in patients in remission and treated with antidepressants found a clear benefit in continuing treatment for up to 1 year for both relapse rate and time to relapse.[19] When discontinuation of treatment is considered, tapering of the antidepressant should be done gradually over several months. Sudden dose reduction or discontinuation of the medication may result in withdrawal syndrome and an increase in anxiety symptoms. For more information on withdrawal syndrome, see Depression.

Following the release of several meta-analyses that demonstrated an increase in suicidal risk associated with the use of antidepressants, many countries including Canada have added a warning about the increased risk of suicidal ideation and behaviour associated with the use of antidepressants in patients <18 years of age.[20][21] However, in light of recent data, the use of antidepressants remains indicated in children and adolescents, but must be prescribed appropriately and combined with strict follow-up. In the adult population, these risks do not seem to be present, as antidepressants have been associated with a protective effect. Close monitoring for suicidal ideation or other indicators of suicidal behaviour remains recommended in this population.[22][23]

Some **benzodiazepines** have demonstrated efficacy in the treatment of anxiety disorders, although they generally do not help with comorbidities like depression.[1][6] They have also been shown to be useful in relieving acute anxiety, agitation and panic attacks at the beginning of treatment while waiting for antidepressant treatment to take effect, as well as to mitigate the temporary increase in anxiety and restlessness that may occur when beginning to take an antidepressant.[1][6] Benzodiazepines remain a second-line choice in the treatment of anxiety disorders because of the risk of abuse, sedation, cognitive impairment, dependence and withdrawal syndrome, and falls associated with their use. It is generally recommended to use them as a short-term measure during the first few weeks of treatment only, if possible. However, benzodiazepines are sometimes continued for a few months if they are effective and well tolerated in the absence of antidepressant response, until an effective treatment is found.[24] In the elderly, benzodiazepines should be used with caution because, due to age-related pharmacokinetic changes, older patients are more likely to experience adverse effects, such as psychomotor and cognitive impairment, and present a higher risk of falls and fractures. Benzodiazepines are not recommended in the presence of a comorbid substance use disorder or a history of one.

Other pharmacological agents include calcium channel modulators such as pregabalin and gabapentin, which have been shown to be effective in the treatment of anxiety disorders in some studies.[1][6] However, there are some case reports of misuse of high-dose pregabalin (including nasal inhalation) in patients with multiple substance use disorders.[25][26] Moreover, a more recent review found that gabapentinoids are sometimes used in combination with certain drugs and that patients with opioid use disorders may be at an increased risk of abusing these types of drugs. Thus, pregabalin should never be prescribed to anyone with a current or past opioid use disorder.[27]

Data are also available on the efficacy of buspirone as well as on the use of mirtazapine, vilazodone, hydroxyzine and trazodone, although the literature on these agents is significantly less comprehensive.

There is evidence to support the use of **second- and third-generation antipsychotics** (**SGAs**) and some **anticonvulsants**, but the limited number of studies as well as the adverse effects profile associated with these agents make them second- or third-line agents or adjunctive treatments.[1][6]

Ultimately, in the patient population receiving treatment, 80% of those with a panic disorder will experience complete remission, while only 50% of those with social phobia will. As for generalized anxiety disorder, when it begins in adolescence, it tends to become chronic, and remission is infrequent. Separation anxiety usually disappears in adulthood.[28]

Psychotherapy or Pharmacotherapy?

Psychotherapy and pharmacotherapy have shown comparable efficacy in the treatment of most anxiety disorders.[13][29] However, in a new meta-analysis conducted with more stringent criteria to assess the efficacy of psychotherapy, the effect size of pharmacotherapy was found to be greater than that of psychotherapy.[30]

The combination of psychotherapy and pharmacotherapy yields different results depending on the anxiety disorder.[6][29][31] Current clinical evidence does not support the combination of psychotherapy and pharmacotherapy in the initial treatment of generalized anxiety disorder. In the treatment of panic disorder, on the other hand, most studies show that the combination of psychotherapy and pharmacotherapy is superior to psychotherapy alone.[6] A study showed that CBT combined with the SSRI sertraline resulted in superior effectiveness, especially with respect to fear of physical sensations, compared to CBT alone.[32] Regarding the treatment of social anxiety disorder, several data on the combination of psychotherapy and medication have been inconclusive.[6]

In the presence of suicidal ideation or treatment-resistant anxiety disorder, pharmacotherapy is generally preferred and a consultation in psychiatry is indicated.

Panic Disorder

Information on the treatment of panic disorder, agoraphobia and panic disorder with agoraphobia comes from studies that did not separate patients with panic disorder and those with agoraphobia into separate groups.[33] The DSM-5-TR draws a distinction between these 2 conditions and panic disorder with agoraphobia, but studies are required to determine whether the pharmacological agents used treat each of these distinct disorders in the same way.

The lifetime prevalence of panic disorder is estimated to be between 4.7–5.1%.[1] It is characterized by recurring and unexpected panic attacks. A panic attack is a sudden surge of intense anxiety that peaks in minutes with the onset of symptoms such as palpitations, tremors, sensation of choking, chest pain or nausea. At least 1 of the panic attacks is followed by persistent fear or worry for a month or more of having other attacks, of attack consequences or behavioural changes related to attacks, such as avoidance related to agoraphobia. The prevalence of panic attacks is higher than that of panic disorder, with up to 40% of the population having a panic attack at some point in their lives.[1]

All SSRIs have demonstrated efficacy in treating panic disorder with or without agoraphobia. In several meta-analyses, SSRIs have been shown to improve panic symptoms, avoidance behaviours associated with agoraphobia, depressive symptomatology and general anxiety.[34][35][36] However, no SSRI has proved superior to any other, although their tolerance varies between individuals and different SSRIs.[37][38][39][40] **Venlafaxine**, an SNRI, has also demonstrated efficacy in reducing the severity of panic disorder symptoms and is associated with a response rate (75%) similar to SSRIs.[41][42]

There is clinical evidence supporting the use of TCAs, including **imipramine** and **clomipramine**, in the treatment of panic disorder. Despite having efficacy similar to SSRIs, TCAs remain second-line agents because of their significant adverse effects profile and the risk of toxicity in overdose.[34]

Benzodiazepines are also a second-line option in the treatment of panic disorder, and **alprazolam**, **clonazepam**, **lorazepam** and **diazepam** have been studied for this indication.[37] However, since these drugs are associated with a risk of abuse, dependence, withdrawal syndrome, falls and CNS adverse effects, they are generally not used for long-term treatment of panic disorder. Their use is mainly limited to the first few weeks of treatment when rapid relief of anxiety or panic attacks is necessary or to reduce the exacerbation of anxiety and agitation that may be present at the beginning of antidepressant treatment.[1] Clonazepam and lorazepam are generally the preferred agents, and while alprazolam and diazepam have a faster onset of action, they present a greater risk of abuse. In addition, with its short elimination half-life, alprazolam is associated with increased anxiety between doses and a greater withdrawal syndrome.[43]

With only one study demonstrating efficacy in the treatment of panic disorder, the use of the TCA desipramine has been relegated to third line.

According to one study, the MAOI **phenelzine** is effective in the treatment of panic disorder, but remains rarely used in practice because of its adverse effect profile, the necessary dietary restrictions to avoid hypertensive crises and the numerous potential drug interactions.[44]

Given the lack of evidence, mirtazapine and moclobemide cannot be recommended in the treatment of panic disorder. Buspirone, trazodone and propranolol are deemed ineffective for this indication.[6]

Table 3 shows the primary recommendations for the pharmacological treatment of panic disorder based on recent evidence.

Table 3: *Overview of Pharmacotherapy for Panic Disorder*

First line	Citalopram, escitalopram, fluoxetine, fluvoxamine, paroxetine, sertraline
	Venlafaxine
Second line	Imipramine, clomipramine
	Alprazolam, clonazepam, lorazepam, diazepam[a]

Third line	Phenelzine
	Desipramine

[a] A benzodiazepine may be used in combination with an antidepressant during the first few weeks of treatment before the onset of action of the antidepressant.

Note: The starting dose of an antidepressant used in the treatment of panic disorder should be as low as possible since patients suffering from this disorder have increased sensitivity to adverse effects that may occur during the initiation of an antidepressant. Therefore, they are more likely to experience an exacerbation of anxiety and agitation at the beginning of treatment, which may lead to premature discontinuation of the medication.

Agoraphobia and Panic Disorder with Agoraphobia

Agoraphobia affects about 1.4% of the population.[1] Patients suffer intense fear and anxiety from not being able to escape or be rescued from at least 2 or more specific situations such as public places, open spaces, crowds and queues. Although the prevalence of agoraphobia is higher with panic disorder, it can also occur alone.

The pharmacologic treatment of agoraphobia with or without panic disorder is the same as for panic disorder. Much of the disability in agoraphobia arises from avoidance behaviour rather than panic attacks. Since the avoidance behaviour is generally not reported by people who have this behaviour, careful questioning or standardized questionnaires may be needed for a complete assessment. The avoidance behaviour can be addressed with **CBT**, since medication is not very effective even if it reduces or eliminates the accompanying panic attacks.

Social Anxiety Disorder (Social Phobia)

Social anxiety disorder is the most common anxiety disorder with a lifetime prevalence of about 8–12%.[1] This excessive fear of being criticized or negatively evaluated by others presents as shyness, avoidance of social contact or difficulty dealing with authority figures. The disorder may be present from childhood, but often becomes noticeable in adolescence.

SSRIs and SNRIs are the treatment of choice for social anxiety disorder. These agents are also effective for social anxiety related to performance. **Escitalopram**, **fluvoxamine**, **paroxetine**, **sertraline** and **venlafaxine** have all demonstrated efficacy.[45][46][47] Results from studies with **fluoxetine** have been inconclusive.[45]

Pregabalin has been shown to be effective in some studies, but only at a high dose (≥600 mg/day) and not at a lower dose (150–300 mg/day).[45][48][49] When used at high doses, pregabalin leads to an increased risk of adverse effects, including cognitive impairment. Since there are no studies comparing the efficacy of pregabalin versus SSRIs or SNRIs in the treatment of social anxiety disorder and antidepressants have the advantage of concomitantly treating a broader range of commonly present comorbid conditions, pregabalin is a second-line option.

As in panic disorder, benzodiazepines, especially **clonazepam** and **bromazepam**, have demonstrated efficacy in the treatment of social anxiety disorder.[45] However, the risks associated with their use are identical and the same recommendations apply as to the duration of their use.

Other second-line agents include **phenelzine**, which has been shown to be effective in several studies; **citalopram**, for which only one study has been conducted; **moclobemide**, which has been shown to be more effective than placebo in a meta-analysis despite several inconclusive data; as well as **mirtazapine** and **gabapentin** for which the efficacy data are limited to one study each.[6][45][50][51]

While only one study has shown the efficacy of **ketamine** in the treatment of social anxiety disorder, its limited availability limits its use and makes it a third-line treatment at this time.

Buspirone[52] and **desvenlafaxine** have been shown to be ineffective in the treatment of social anxiety disorder.

Table 4 shows the primary recommendations for the pharmacological treatment of social anxiety disorder based on recent evidence.

Table 4: *Overview of Pharmacotherapy for Social Anxiety Disorder*

First line	Escitalopram, fluvoxamine, paroxetine, sertraline
	Venlafaxine
Second line	Pregabalin, gabapentin
	Bromazepam, clonazepam[a]
	Phenelzine
	Moclobemide
	Citalopram
	Mirtazapine
Third line	Ketamine

[a] A benzodiazepine may be used in combination with an antidepressant during the first few weeks of treatment before the onset of action of the antidepressant.

Note: A low dose of **propranolol** or **atenolol** taken 30–60 minutes before an anxiety-provoking event may reduce stage fright or fear of public speaking, but these drugs remain ineffective in treating generalized social anxiety disorder when not related to public performance.[53][54][55]

Specific Phobia

Specific phobia is characterized by intense fear or anxiety associated with certain objects or situations, frequently leading to avoidance behaviours. The most common types of phobia involve animals, natural disasters, certain environments (enclosed spaces, airplanes), injections, injuries or blood.

This condition occurs in approximately 10–13% of the population over the course of a lifetime and is more prevalent among adolescents (up to 36.5%).[1] The onset of specific phobia occurs at around 5–12 years of age, although this may vary depending on the type of phobia.[1]

Exposure therapy is the treatment of choice for specific phobia and is often enough to produce a marked and lasting improvement.[56] Pharmacotherapy is rarely needed and there are little data on it. However, studies have shown that the use of a pharmacologic agent, including **alprazolam** or **pregabalin**, taken occasionally before exposure may be helpful.[6][57][58]

Fluoxetine has been shown to be effective in a study conducted with patients with various specific phobias.[6]

Generalized Anxiety Disorder

Generalized anxiety disorder (GAD) is characterized by excessive and uncontrollable worry related to everyday life concerns such as safety of family members, financial/job security and health occurring most of the time for ≥6 months. **CBT** is the most effective psychosocial treatment but often requires 20 or more sessions to be successful.

The lifetime prevalence of generalized anxiety disorder is approximately 6%.[1] As with other anxiety disorders, this condition affects women more than men. It is also the most common anxiety disorder in people >65 years of age.[1] GAD is not always easily diagnosed, so only one-third of those affected are adequately treated.[1]

With its vague symptomatology, generalized anxiety disorder is the anxiety disorder that occurs most often in association with other psychiatric comorbidities, including depression and other anxiety disorders, as well as physical disorders, including pain syndrome, hypertension, cardiovascular diseases, and gastrointestinal disorders.[1]

SSRIs and SNRIs are established as first-line treatment for generalized anxiety disorder with comparable efficacy. Numerous studies have demonstrated the efficacy of **escitalopram**,[59][60][61] **paroxetine**,[62][63][64] **sertraline**,[65][66] **venlafaxine** and **duloxetine**.[67][68][69][70] In a study conducted among the elderly, **citalopram** was proven to be superior to placebo.

Several pharmacological agents can be used as second-line treatment for generalized anxiety disorder.

The efficacy of **vilazodone** in the treatment of generalized anxiety disorder has been shown. However, there have been no studies comparing it with active treatment, nor any long-term studies.[6]

Although several studies support the use of **pregabalin** for generalized anxiety disorder, the risk of abuse is a growing concern, making it a second-line treatment.[6][71][72][73][74]

Alprazolam, **bromazepam**, **diazepam** and **clonazepam** are benzodiazepines that have demonstrated efficacy in the treatment of generalized anxiety disorder. They are nonetheless second-line choices with the same limitations associated with their use as in the treatment of panic disorder or social anxiety.[75]

Although effective in generalized anxiety disorder, the use of **imipramine** is limited by its adverse effects and the dangers it presents in the case of overdose.[76][77]

A 12-week study has shown **bupropion** to have a comparable efficacy to escitalopram.[78]

Significant evidence supports the efficacy of **quetiapine** XR as monotherapy for generalized anxiety disorder.[79][80] At a dose of 50–150 mg daily, it has been shown to rapidly reduce many symptoms of generalized anxiety disorder. However, quetiapine is considered a third-line choice due to its significant sedative effect, potential for weight gain and potential effects on metabolic regulation.

Hydroxyzine has been shown to be as effective as benzodiazepines and buspirone in a single randomized controlled trial.[81] However, clinical use of this agent remains very limited due to its adverse effect profile, and it is not typically included in current pharmacotherapy for generalized anxiety disorder.

There is limited evidence of the efficacy of **trazodone** and **valproate** in the treatment of generalized anxiety disorder.

A few studies have been conducted with second-generation antipsychotics as adjuvant therapy for generalized anxiety disorder. **Olanzapine** has demonstrated efficacy in combination with an antidepressant,[82] but given its adverse effect profile, it should be considered only as an adjuvant in refractory cases.

Buspirone has shown comparable efficacy to benzodiazepines with lower sedative potential and a very low risk of abuse.[83] Like antidepressants, it has a slow onset of action of a few weeks. However, few studies have compared the effectiveness of buspirone to antidepressants. Data on long-term use and on relapse prevention are lacking. Buspirone is thus infrequently used in practice.

Table 5 shows the primary recommendations for the pharmacological treatment of generalized anxiety disorder based on recent evidence.

***Table 5:** Overview of Pharmacotherapy for Generalized Anxiety Disorder*

First line	Escitalopram, paroxetine, sertraline
	Venlafaxine, duloxetine
Second line	Vilazodone
	Pregabalin
	Alprazolam, bromazepam, clonazepam, diazepam
	Imipramine
Third line	Quetiapine

	Hydroxyzine
	Trazodone
	Valproate
Refractory GAD	Olanzapine + antidepressant
	Pregabalin + antidepressant

Choices during Pregnancy and Breastfeeding

Anxiety Disorders and Pregnancy

Anxiety disorders have onset early in life, are frequently chronic in nature, and their severity waxes and wanes in response to environmental events, but prevalence remains unchanged during pregnancy. Anxious disorders therefore occur in approximately 15–20% of patients during pregnancy and the postpartum period.[84][85] A decrease or increase in anxiety can occur during pregnancy. Childbirth with significant complications can also lead to PTSD.

Anxiety disorders and associated comorbidities can have a negative impact during pregnancy on the patient as well as on the developing child. A meta-analysis found no link between anxiety symptoms and perinatal outcome,[86] but more recent meta-analyses[87][88][89] report that anxiety disorders may be associated with preterm birth or low birth weight. Anxiety symptoms during pregnancy were also associated with depressive symptoms, substance abuse and decreased prenatal vitamin adherence, as well as an increased choice of cesarean delivery. Parental skills can also be affected by anxiety disorders. Patients with an anxiety disorder are less likely to promote their child's psychological independence.[90]

It is therefore important to screen for the presence of anxiety symptoms prior to conception, if possible. Screening can be repeated during pregnancy and especially postpartum. Preconception treatment may be offered for anxiety disorders that are causing significant distress or interfering with normal functioning. Panic disorder with agoraphobia may prevent a patient from attending medical appointments because of phobic avoidance. In a patient with severe anxiety related to pregnancy, postpartum and breastfeeding, it is essential to look for the symptoms of mood disorders, suicidal ideation and thoughts of infanticide.

In any circumstance where a patient experiences severe anxiety or mood disorder symptoms during pregnancy or postpartum, referral to a psychiatrist is necessary. In major centres, mental health programs are usually available and are attuned to responding to consultation requests quickly.

Management during Pregnancy

There is evidence that psychological treatments can have beneficial effects for more than half of the patients who persist with a treatment program even if data specific to pregnancy are not available. **CBT** and **interpersonal therapy (IPT)** are recommended as first-line treatment for mild to moderate anxiety disorders.[91] Therapies based on meditative or relaxation techniques have also demonstrated efficacy and may be more acceptable for pregnant patients than pharmacologic approaches.[92][93]

If anxiety symptoms are severe and cause significant impairment or distress, pharmacotherapy can be appropriate and effective in pregnant patients.[94] The 2 main classes of medications used for anxiety disorders during pregnancy are SSRI or SNRI antidepressants and benzodiazepines.

The use of **SSRIs** during the first trimester of pregnancy has been linked to spontaneous abortions, although no risk factors have been clearly identified.[95] Studies conducted with large populations controlling for confounding factors suggest that SSRI exposure during the first trimester is not associated with major congenital anomalies.[95] However, congenital cardiac abnormalities have been associated with paroxetine exposure, so it is recommended to avoid this medication during pregnancy.[95][96] In the third trimester, SSRIs may lead to behavioural syndrome of the newborn that is characterized by symptoms such as tremors, agitation, increased muscle tone, digestive or nutritional disorders, breathing difficulties, and sometimes convulsions.[95] Paroxetine and fluoxetine have been involved more frequently in the onset of this syndrome, possibly because these molecules cross the placental barrier more than other SSRIs.[95][96][97] Finally, the results of studies on the association between SSRI exposure in utero and the development of autism spectrum disorder remain contradictory.[95]

There are limited data on the use of **SNRIs** during pregnancy, but their use does not appear to be associated with an increased risk of major congenital malformations. The use of these drugs during the third trimester of pregnancy, however, carries a risk of behavioural syndrome of the newborn.[94]

Multiple reviews and meta-analyses concluded that the use of **benzodiazepines** during pregnancy does not increase the risk of congenital malformations, but the association identified in the case-control studies can not be ruled out. It is therefore recommended to avoid their use during the first trimester of pregnancy.[91][98] In a recent meta-analysis, prenatal exposure to benzodiazepines was reported to be associated with increased risks of several perinatal adverse events, including spontaneous miscarriage, preterm birth, low birth weight, low Apgar score and admission to the neonatal intensive care unit.[84] However, the authors identified several potential confounding factors, including the disease for which the benzodiazepines were prescribed, such as epilepsy, obstetric use of benzodiazepines or a psychiatric disorder that may induce adverse perinatal events, as well as tobacco or illicit substance use. They therefore concluded that patients receiving benzodiazepines should be monitored for high-risk pregnancy. In addition, a first systematic review focussed on the combination of benzodiazepines and antidepressants. This study found that using this combination during the first trimester is associated with a risk of major non-specific malformations, but since the relative risk is statistically low, the clinical significance is also low.[99] Nonetheless, the authors recommend avoiding polypharmacy, since the use of benzodiazepines or antidepressants as monotherapy does not appear to be associated with teratogenicity.

The use of SSRIs, SNRIs or benzodiazepines may be totally warranted in patients for whom the severity of symptoms could affect fetal or maternal safety or health. The lowest effective dose should then be used.[95] If an antidepressant was already taken for an anxiety disorder before pregnancy, it can be continued at the same dose.[95] If antidepressant treatment is to be initiated during pregnancy, citalopram, escitalopram and sertraline are the preferred agents in light of the latest evidence, and paroxetine should be avoided because of its association with a risk of congenital cardiac malformations.[95][97] General principles for management of depression during pregnancy are applicable to the management of anxiety disorders.[100] See also Depression for information on management of depression during pregnancy and breastfeeding.

Anxiety Disorders and Breastfeeding

In the postpartum period, severe anxiety with or without depressive symptoms can impede the patient's sleep and erode their confidence in caring for the child. If a patient is unable or refuses to be involved with caring for the child, an urgent psychiatric consultation is needed and treatment considerations should include admitting the patient and child to the hospital.

As in pregnancy, nonpharmacologic options should be used whenever possible in the postpartum period, particularly in breastfeeding patients. If drug therapy is necessary, consider **paroxetine** or **sertraline**, since low amounts are transferred into breast milk and they are undetected in the plasma of the breastfed child.[95][97][101] Avoid **benzodiazepines** due to potential accumulation, sedation and impaired temperature regulation in the infant.

A discussion of general principles on the use of medications in these special populations can be found in Drugs Use during Pregnancy and Drug Use during Breastfeeding. Other specialized reference sources are also provided in these appendices.

Treatment of Anxiety Disorders in Children and Adolescents

Some fears arise transiently during a child's development and do not need to be treated, such as fear of the dark, strangers or heights. However, anxiety disorders are among the most common psychiatric disorders in children and adolescents, with a lifetime prevalence in the United States of 20–30%.[102] The average age of onset for anxiety disorders is 11 years old; separation anxiety, specific phobias and social anxiety disorder appear in childhood or early adolescence, while panic disorder, agoraphobia and generalized anxiety disorder generally appear later. Untreated anxiety disorders can have serious consequences, including impaired social and academic development and functioning, and lead to other psychiatric disorders in adulthood, such as depression and substance abuse. Panic disorder or generalized anxiety disorder combined with depression is the most significant risk factor for developing suicidal ideation or behaviour in adolescents.[103]

CBT can be considered a first-line treatment for children 6–18 years of age, especially when the anxiety disorder is mild to moderate. **SSRIs** and possibly **SNRIs** are treatment options in more severe cases or when CBT is not available. The combination of CBT and an antidepressant may be more effective than either treatment alone in treating social anxiety disorder, generalized anxiety disorder, separation anxiety, or panic disorder, or when the anxiety disorder is severe or incapacitating, when there is an urgency to treat, or when a partial response is obtained with monotherapy.[103]

The SSRIs for which there is sufficient evidence are **fluoxetine**, **fluvoxamine**, **paroxetine** and **sertraline**. There are no SSRIs indicated for the treatment of anxiety disorders in children and adolescents in Canada or the US, and they all carry a warning about the risk of suicidal ideation and suicidal behavior (up to age 18 in Canada and 24 in the US). This risk is estimated at 0.7%, with a number needed to harm (NNH) of 143 compared with a number needed to treat (NNT) of 3.[104] There are fewer empirical data for SNRIs than for SSRIs in children and adolescents; the agents for which there are sufficient data to justify their use are **duloxetine** and **venlafaxine**.

Duloxetine is the only antidepressant with an official indication in the US (not Canada) for the treatment of anxiety in children and adolescents, specifically for generalized anxiety disorder in children and adolescents 7–17 years of age. In children and adolescents, venlafaxine is potentially associated with a greater risk of suicide than other pharmacological agents,[105][106] and both venlafaxine and desvenlafaxine have been associated with fatal overdoses.[103]

Prior to initiating treatment, a comprehensive assessment is required, including a meeting with the child and a parent/caregiver, as well as a differential diagnosis to rule out physical conditions such as hyperthyroidism, migraine, asthma, diabetes, central nervous system disorders, cardiac disorders, systemic lupus erythematosus, pheochromocytoma, lead poisoning and dysmenorrhea. Differential diagnosis also includes ruling out attention deficit hyperactivity disorder (ADHD), bipolar disorder, OCD, psychosis and autism spectrum disorder. Likewise, the use of medications that can cause anxiety, caffeine, tobacco, over-the-counter products, drugs and alcohol must be investigated. Child and caregiver psychoeducation is essential to ensuring good adherence and monitoring of possible adverse effects.

Therapeutic Tips

- Short-term interventions such as relaxation and problem-solving techniques may be helpful.
- Benzodiazepines used for a limited period of time, e.g., up to 6–8 weeks, may also be effective in providing rapid relief from anxiety or panic attacks when needed or to help reduce anxiety and agitation related to initiation of SSRIs and SNRIs.
- Assess a drug's effectiveness after a trial of an adequate dosage taken for a sufficient length of time.
- The use of self-reported normalized scales can facilitate monitoring the course of the anxiety disorder as well as assess treatment effectiveness.
- If the first antidepressant at optimal dose is not effective or not tolerated, it is recommended to switch to another first-line agent.
- When there is partial response to a first-line agent at optimal dose (e.g., some benefit), augmentation with another agent is recommended.

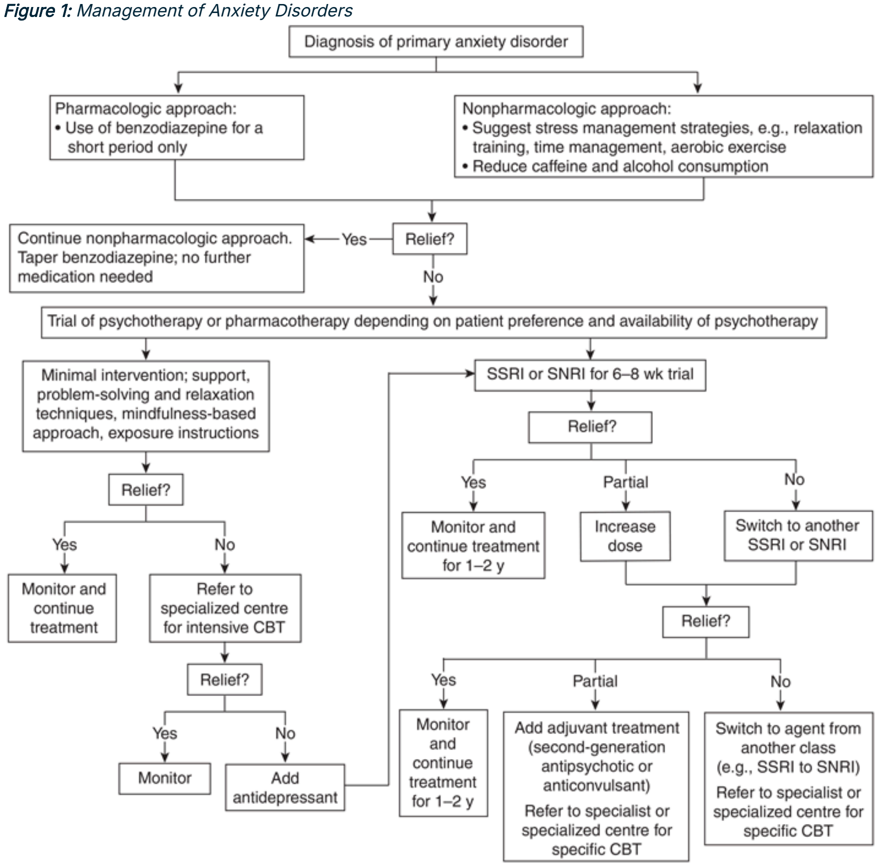
Resources

The following resources can be useful and suggested to patients:

- The Mindshift CBT app is available through Anxiety Canada.
- Phobies-Zéro (French only) offers support for people with anxiety disorders and their families and caregivers, including self-management workshops for anxiety disorders, information and support groups.
- Relief offers support groups and self-management workshops to help people living with anxiety, depression or bipolarity.

Algorithms

Figure 1: Management of Anxiety Disorders



Abbreviations: CBT = cognitive behavioural therapy; SNRI = serotonin-norepinephrine reuptake inhibitor; SSRI = selective serotonin reuptake inhibitor

Drug Table

***Table 6:** Drugs Used for the Management of Anxiety Disorders*

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
Drug Class: Antipsychotics, second–generation					
quetiapine Seroquel, Seroquel XR, generics < $40	GAD	Initial: 50 mg daily PO Target: 50–150 mg daily PO	CNS: drowsiness, psychomotor impairment, headache. Anticholinergic: dry mouth, constipation, blurred vision, urinary retention, cognitive impairment. Cardiovascular: orthostatic hypotension, palpitations, dizziness, tachycardia. Others: weight gain, metabolic disorders (dyslipidemia, diabetes).	Metabolized by CYP3A4. Caution with drugs that may alter quetiapine's metabolism. Increased risk of additive adverse effects and respiratory depression when combined with other CNS depressants such as alcohol.	Considering the adverse effects of quetiapine, monitor anthropometric and metabolic parameters.
Drug Class: Azapirones					
buspirone generics < $40–80	GAD	Initial: 5 mg BID–TID PO Titrate gradually to effective dose Target: 10–60 mg/day PO	CNS: fatigue, headache, agitation (especially on initiation of therapy). Others: nausea, dizziness.	Metabolized by CYP3A4. Caution with drugs that may alter buspirone's metabolism. Avoid combination with MAOI due to an increased risk of serotonin syndrome.	Onset of action approximately 2–4 wk. Should not be used PRN.

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
Drug Class: Benzodiazepines					
clonazepam Rivotril, generics < $40	AG, GAD, PD, SAD	Initial: 0.25–0.5 mg BID PO Target: 1–2 mg daily PO	CNS: drowsiness, psychomotor impairment, fatigue, muscle weakness, reduced concentration, confusion, dysarthria, ataxia, paradoxical agitation (rare), retrograde amnesia (especially with high doses).	Increased risk of additive adverse effects and respiratory depression when combined with other CNS depressants such as alcohol.	Tolerance to drowsiness develops with chronic use. Abuse potential, particularly those with rapid onset of action such as alprazolam and diazepam. Risk of addiction. Withdrawal syndrome following abrupt discontinuation of treatment may precipitate a return of initial symptoms or the emergence of rebound symptoms with intensification of initial symptoms or withdrawal with emergence of other symptoms such as nausea, tremors and seizures. Use with caution in the elderly as they may be more likely to experience adverse effects such as psychomotor and cognitive impairment and impaired coordination with risk of falling. Avoid in patients with a current or a history of substance use disorder.
lorazepam Ativan, generics < $40	AG, GAD, PD	Initial: 0.5 mg TID-QID PO Target: 2–8 mg daily PO	CNS: drowsiness, psychomotor impairment, fatigue, muscle weakness, reduced concentration, confusion, dysarthria, ataxia, paradoxical agitation (rare), retrograde amnesia (especially with high doses).	Increased risk of additive adverse effects and respiratory depression when combined with other CNS depressants such as alcohol.	Tolerance to drowsiness develops with chronic use. Abuse potential, particularly those with rapid onset of action such as alprazolam and diazepam. Risk of addiction. Withdrawal syndrome following abrupt discontinuation of treatment may precipitate a return of initial symptoms or the emergence of rebound symptoms with intensification of initial symptoms or withdrawal with emergence of other symptoms such as nausea, tremors and seizures. Use with caution in the elderly as

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
					they may be more likely to experience adverse effects such as psychomotor and cognitive impairment and impaired coordination with risk of falling. Avoid in patients with a current or a history of substance use disorder. Lorazepam is a good option in hepatic impairment.
Drug Class: Beta$_1$-adrenergic Antagonists					
atenolol Tenormin, generics < $40	Specific task-related anxiety	Initial: 25 mg PO 60 min before task PRN Target: 25–50 mg PO	Hypotension, bradycardia.		For occasional use in specific situations such as public speaking or performing; not useful in social anxiety disorder when not related to public performance.
propranolol generics < $40	Specific task-related anxiety	Initial: 10 mg PO 30–60 min before task PRN Target: 10–40 mg PO	Hypotension, bradycardia.		For occasional use in specific situations such as public speaking or performing; not useful in social anxiety

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
					disorder when not related to public performance.
Drug Class: Dual Action Antidepressants					
mirtazapine Remeron, Remeron RD, generics < $40	SAD	Initial: 15 mg/day PO Target: 15–60 mg/day PO	CNS: fatigue, somnolence. GI: constipation, increased risk of upper GI bleeding. Cardiovascular: orthostatic hypotension, palpitations, dizziness, QT_c interval prolongation. Others: dry mouth, increased appetite and weight gain, hypertriglyceridemia, SIADH with hyponatremia.	Mirtazapine is metabolized by the liver. Caution with drugs that may alter its metabolism. Combination with an MAOI is contraindicated due to an increased risk of serotonin syndrome. Caution with other serotonergic drugs such as amphetamine derivatives, dextromethorphan, meperidine, St. John's wort, tryptophan, etc., due to risk of serotonin syndrome. Increased risk of GI bleeding with NSAIDs or antiplatelet agents. Consider other risk factors for GI bleeding.	Due to its greater antihistamine effect, low-dose mirtazapine is more likely to cause drowsiness. Mirtazapine causes minimal sexual dysfunction.

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
Drug Class: Gabapentinoids					
gabapentin Neurontin, generics < $40	SAD	Initial: 300 mg/day PO Target: 900–3600 mg/day PO in divided doses	CNS: drowsiness, fatigue, dizziness, ataxia, impaired coordination, impaired vision. Others: peripheral edema.	Magnesium- and aluminum-containing antacids may decrease the absorption of gabapentin. Increased risk of additive adverse effects and respiratory depression with concurrent use of other CNS depressants such as alcohol and opioids.	Withdrawal symptoms may include nausea, diarrhea, headaches, nightmares and other sleep disorders, paresthesias, diaphoresis, muscle spasms, tachycardia and hallucinations.
pregabalin Lyrica, generics < $40	GAD, SAD	Initial: 50 mg TID PO GAD target: 150–600 mg daily PO in divided doses SAD target: ≥600 mg daily PO in divided doses	CNS: drowsiness, fatigue, dizziness, ataxia, impaired coordination, impaired vision, headache, tremors. Others: nausea, dry mouth, peripheral edema.	Increased risk of additive adverse effects and respiratory depression with concurrent use of other CNS depressants such as alcohol and opioids.	Withdrawal symptoms may include nausea, diarrhea, headaches, nightmares and other sleep disorders, paresthesias, diaphoresis, muscle spasms, tachycardia and hallucinations. A risk of abuse has been reported, particularly in people with substance use disorders, with opioid use disorder representing the greatest risk factor for gabapentinoid abuse. The use of pregabalin in patients with substance use disorders is not recommended.
Drug Class: Monoamine Oxidase Inhibitors (MAOIs), irreversible					
phenelzine Nardil < $40–80	AG, PD, SAD	Initial: 15 mg daily PO Target: 15–90 mg daily PO	CNS: drowsiness, fatigue, insomnia, headache, tremors, seizure. Anticholinergic: dry mouth, constipation, blurred vision, urinary retention, cognitive impairment. Cardiovascular: orthostatic hypotension, palpitations, dizziness, tachycardia. Others: increased appetite, weight gain, sexual dysfunction.	Concurrent use with sympathomimetic agents, tyramine or levodopa may result in hypertensive crisis; do not use with serotonergic drugs such as SSRI, SNRIs, TCA, triptans, dextromethorphan, meperidine and tryptophan due to high risk of serotonin syndrome.	MAOIs are toxic in overdose; contraindicated in case of suicidality. Use with caution in case of cardiovascular disease, cognitive impairment or history of seizure. Stringent dietary restrictions are necessary. Avoid tyramine-containing foods. Contraindicated if ClCr <30 mL/min.
Drug Class: Selective Serotonin Reuptake Inhibitors (SSRIs)					
citalopram Celexa, CTP 30, generics < $40	AG, PD, GAD, SAD	Initial: 10 mg/day PO Target: 20–40 mg/day PO	CNS: anxiety, agitation, insomnia, headache, extrapyramidal effects. GI: nausea, vomiting, diarrhea,	SSRI are all metabolized by the liver. Caution with drugs that may alter their metabolism.	Adverse effect profile varies depending on the SSRI. Citalopram and escitalopram: SSRI exhibiting

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
		Maximum: 40 mg/day PO and 20 mg/day PO in ≥65 y	constipation, increased risk of upper GI bleeding. Others: dry mouth, increased sweating, sexual dysfunction, SIADH with hyponatremia.	All SSRI do not have the same potential for drug interactions. Citalopram and escitalopram: low enzyme inhibitors. Fluoxetine and paroxetine: potent CYP2D6 inhibitors. Fluvoxamine: potent CYP1A2 and other CYPs inhibitors. Sertraline: weak CYP2D6 inhibitor. Combination with MAOI is contraindicated due to an increased risk of serotonin syndrome. Caution with other serotonergic drugs such as amphetamine derivatives, dextromethorphan, meperidine and St. John's wort due to an increased risk of serotonin syndrome. Increased risk of GI bleeding with the combination of an SSRI and an NSAID or an	the highest selectivity. May be better tolerated. Health Canada–issued recommendations including maximum doses (see Dosage) based upon the risk of QT_c prolongation with these agents. Fluoxetine: stimulating, frequent sexual dysfunction. Fluvoxamine: highly sedating, significant GI disturbances. Paroxetine: somewhat sedating, anticholinergic effects, significant sexual disturbances and weight gain. Sertraline: somewhat stimulating; take with food to increase absorption.

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
				antiplatelet agent. Consider other risk factors for GI bleeding. SSRI may prolong QT_c interval, especially citalopram and escitalopram. Use with caution with other drugs associated with prolonged QT_c interval or other risk factors.	
escitalopram Cipralex, KYE-Escitalopram, other generics < $40	AG, GAD, PD, SAD	Initial: 5 mg/day PO Target: 10–20 mg/day PO Maximum: 20 mg/day PO and 10 mg/day PO in ≥65 y	CNS: anxiety, agitation, insomnia, headache, extrapyramidal effects. GI: nausea, vomiting, diarrhea, constipation, increased risk of upper GI bleeding. Others: dry mouth, increased sweating, sexual dysfunction, SIADH with hyponatremia.	SSRI are all metabolized by the liver. Caution with drugs that may alter their metabolism. All SSRI do not have the same potential for drug interactions. Citalopram and escitalopram: low enzyme inhibitors. Fluoxetine and paroxetine: potent CYP2D6 inhibitors. Fluvoxamine: potent CYP1A2 and other CYPs inhibitors. Sertraline: weak CYP2D6 inhibitor.	Adverse effect profile varies depending on the SSRI. Citalopram and escitalopram: SSRI exhibiting the highest selectivity. May be better tolerated. Health Canada–issued recommendations including maximum doses (see Dosage) based upon the risk of QT_c prolongation with these agents. Fluoxetine: stimulating, frequent sexual dysfunction. Fluvoxamine: highly sedating,

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
				Combination with MAOI is contraindicated due to an increased risk of serotonin syndrome. Caution with other serotonergic drugs such as amphetamine derivatives, dextromethorphan, meperidine and St. John's wort due to an increased risk of serotonin syndrome. Increased risk of GI bleeding with the combination of an SSRI and an NSAID or an antiplatelet agent. Consider other risk factors for GI bleeding. SSRI may prolong QT_c interval, especially citalopram and escitalopram. Use with caution with other drugs associated with prolonged QT_c interval or other risk factors.	significant GI disturbances. Paroxetine: somewhat sedating, anticholinergic effects, significant sexual disturbances and weight gain. Sertraline: somewhat stimulating; take with food to increase absorption.
fluoxetine Prozac, Odan-Fluoxetine, other generics < $40	AG, PD	Initial: 10 mg/day PO Target: 20–60 mg/day PO	CNS: anxiety, agitation, insomnia, headache, extrapyramidal effects. GI: nausea, vomiting, diarrhea, constipation, increased risk of upper GI bleeding. Others: dry mouth, increased sweating, sexual dysfunction, SIADH with hyponatremia.	SSRI are all metabolized by the liver. Caution with drugs that may alter their metabolism. All SSRI do not have the same potential for drug interactions. Citalopram and escitalopram: low enzyme inhibitors. Fluoxetine and paroxetine: potent CYP2D6 inhibitors. Fluvoxamine: potent CYP1A2 and other CYPs inhibitors. Sertraline: weak CYP2D6 inhibitor. Combination with MAOI is contraindicated due to an increased risk of serotonin syndrome. Caution with other serotonergic drugs such as amphetamine derivatives, dextromethorphan, meperidine and St. John's wort due to an increased risk of serotonin syndrome.	Adverse effect profile varies depending on the SSRI. Citalopram and escitalopram: SSRI exhibiting the highest selectivity. May be better tolerated. Health Canada–issued recommendations including maximum doses (see Dosage) based upon the risk of QT_c prolongation with these agents. Fluoxetine: stimulating, frequent sexual dysfunction. Fluvoxamine: highly sedating, significant GI disturbances. Paroxetine: somewhat sedating, anticholinergic effects, significant sexual disturbances and weight gain. Sertraline: somewhat stimulating; take with food to increase absorption.

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
				Increased risk of GI bleeding with the combination of an SSRI and an NSAID or an antiplatelet agent. Consider other risk factors for GI bleeding. SSRI may prolong QT_c interval, especially citalopram and escitalopram. Use with caution with other drugs associated with prolonged QT_c interval or other risk factors.	
fluvoxamine Luvox, generics < $40	AG, PD, SAD	Initial: 25 mg/day PO Target: 100–300 mg/day PO	CNS: anxiety, agitation, insomnia, headache, extrapyramidal effects. GI: nausea, vomiting, diarrhea, constipation, increased risk of upper GI bleeding. Others: dry mouth, increased sweating, sexual dysfunction, SIADH with hyponatremia.	SSRI are all metabolized by the liver. Caution with drugs that may alter their metabolism. All SSRI do not have the same potential for drug interactions. Citalopram and escitalopram: low enzyme inhibitors. Fluoxetine and paroxetine: potent CYP2D6 inhibitors. Fluvoxamine: potent	Adverse effect profile varies depending on the SSRI. Citalopram and escitalopram: SSRI exhibiting the highest selectivity. May be better tolerated. Health Canada–issued recommendations including maximum doses (see Dosage) based upon the risk of QT_c prolongation with these agents.

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
				CYP1A2 and other CYPs inhibitors. Sertraline: weak CYP2D6 inhibitor. Combination with MAOI is contraindicated due to an increased risk of serotonin syndrome. Caution with other serotonergic drugs such as amphetamine derivatives, dextromethorphan, meperidine and St. John's wort due to an increased risk of serotonin syndrome. Increased risk of GI bleeding with the combination of an SSRI and an NSAID or an antiplatelet agent. Consider other risk factors for GI bleeding. SSRI may prolong QT_c interval, especially citalopram and escitalopram. Use with caution with other drugs associated with prolonged QT_c	Fluoxetine: stimulating, frequent sexual dysfunction. Fluvoxamine: highly sedating, significant GI disturbances. Paroxetine: somewhat sedating, anticholinergic effects, significant sexual disturbances and weight gain. Sertraline: somewhat stimulating; take with food to increase absorption.

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
				interval or other risk factors.	
paroxetine immediate-release Paxil, generics < $40	AG, PD, GAD, SAD	Initial: 10 mg/day PO Target: 20–60 mg/day PO	CNS: anxiety, agitation, insomnia, headache, extrapyramidal effects. GI: nausea, vomiting, diarrhea, constipation, increased risk of upper GI bleeding. Others: dry mouth, increased sweating, sexual dysfunction, SIADH with hyponatremia.	SSRI are all metabolized by the liver. Caution with drugs that may alter their metabolism. All SSRI do not have the same potential for drug interactions. Citalopram and escitalopram: low enzyme inhibitors. Fluoxetine and paroxetine: potent CYP2D6 inhibitors. Fluvoxamine: potent CYP1A2 and other CYPs inhibitors. Sertraline: weak CYP2D6 inhibitor. Combination with MAOI is contraindicated due to an increased risk of serotonin syndrome. Caution with other serotonergic drugs such as amphetamine derivatives, dextromethorphan, meperidine and St. John's wort due to an increased risk of serotonin syndrome. Increased risk of GI bleeding with the combination of an SSRI and an NSAID or an antiplatelet agent. Consider other risk factors for GI bleeding. SSRI may prolong QT_c interval, especially citalopram and escitalopram. Use with caution with other drugs associated with prolonged QT_c interval or other risk factors.	Adverse effect profile varies depending on the SSRI. Citalopram and escitalopram: SSRI exhibiting the highest selectivity. May be better tolerated. Health Canada–issued recommendations including maximum doses (see Dosage) based upon the risk of QT_c prolongation with these agents. Fluoxetine: stimulating, frequent sexual dysfunction. Fluvoxamine: highly sedating, significant GI disturbances. Paroxetine: somewhat sedating, anticholinergic effects, significant sexual disturbances and weight gain. Sertraline: somewhat stimulating; take with food to increase absorption.
paroxetine controlled-release Paxil CR $80–160	AG, PD, GAD, SAD	Initial: 12.5 mg/day PO Target: 25–50 mg/day PO	CNS: anxiety, agitation, insomnia, headache, extrapyramidal effects. GI: nausea, vomiting, diarrhea, constipation, increased risk of upper GI bleeding. Others: dry mouth, increased sweating, sexual dysfunction, SIADH with hyponatremia.	SSRI are all metabolized by the liver. Caution with drugs that may alter their metabolism. All SSRI do not have the same potential for drug interactions. Citalopram and escitalopram: low enzyme inhibitors. Fluoxetine and paroxetine: potent	Adverse effect profile varies depending on the SSRI. Citalopram and escitalopram: SSRI exhibiting the highest selectivity. May be better tolerated. Health Canada–issued recommendations including maximum doses (see Dosage) based upon the

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
				CYP2D6 inhibitors. Fluvoxamine: potent CYP1A2 and other CYPs inhibitors. Sertraline: weak CYP2D6 inhibitor. Combination with MAOI is contraindicated due to an increased risk of serotonin syndrome. Caution with other serotonergic drugs such as amphetamine derivatives, dextromethorphan, meperidine and St. John's wort due to an increased risk of serotonin syndrome. Increased risk of GI bleeding with the combination of an SSRI and an NSAID or an antiplatelet agent. Consider other risk factors for GI bleeding. SSRI may prolong QT_c interval, especially citalopram and escitalopram. Use	risk of QT_c prolongation with these agents. Fluoxetine: stimulating, frequent sexual dysfunction. Fluvoxamine: highly sedating, significant GI disturbances. Paroxetine: somewhat sedating, anticholinergic effects, significant sexual disturbances and weight gain. Sertraline: somewhat stimulating; take with food to increase absorption.
				with caution with other drugs associated with prolonged QT_c interval or other risk factors.	
sertraline Zoloft, generics < $40	AG, PD, GAD, SAD	Initial: 25 mg/day PO Target: 50–200 mg/day PO	CNS: anxiety, agitation, insomnia, headache, extrapyramidal effects. GI: nausea, vomiting, diarrhea, constipation, increased risk of upper GI bleeding. Others: dry mouth, increased sweating, sexual dysfunction, SIADH with hyponatremia.	SSRI are all metabolized by the liver. Caution with drugs that may alter their metabolism. All SSRI do not have the same potential for drug interactions. Citalopram and escitalopram: low enzyme inhibitors. Fluoxetine and paroxetine: potent CYP2D6 inhibitors. Fluvoxamine: potent CYP1A2 and other CYPs inhibitors. Sertraline: weak CYP2D6 inhibitor. Combination with MAOI is contraindicated due to an increased risk of serotonin syndrome. Caution with other serotonergic drugs such as	Adverse effect profile varies depending on the SSRI. Citalopram and escitalopram: SSRI exhibiting the highest selectivity. May be better tolerated. Health Canada–issued recommendations including maximum doses (see Dosage) based upon the risk of QT_c prolongation with these agents. Fluoxetine: stimulating, frequent sexual dysfunction. Fluvoxamine: highly sedating, significant GI disturbances. Paroxetine: somewhat sedating, anticholinergic effects, significant sexual disturbances and weight gain.

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
				amphetamine derivatives, dextromethorphan, meperidine and St. John's wort due to an increased risk of serotonin syndrome. Increased risk of GI bleeding with the combination of an SSRI and an NSAID or an antiplatelet agent. Consider other risk factors for GI bleeding. SSRI may prolong QT_c interval, especially citalopram and escitalopram. Use with caution with other drugs associated with prolonged QT_c interval or other risk factors.	Sertraline: somewhat stimulating; take with food to increase absorption.
Drug Class: Serotonin Modulators					

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
vilazodone Viibryd $80–120	GAD	Initial: 10 mg/day PO with food Target: 20–40 mg/day PO with food	CNS: insomnia, headache. GI: nausea, vomiting, abdominal pain, diarrhea, increased risk of upper GI bleeding. Others: dry mouth, dizziness, SIADH with hyponatremia.	Vilazodone is metabolized by the liver. Caution with drugs that may alter its metabolism. Maximum dose is 20 mg/day when combined with a strong CYP3A4 inhibitor. It is also a moderate CYP2C19 inhibitor and, to a lesser extent, CYP2D6. Combination with an MAOI is contraindicated due to an increased risk of serotonin syndrome. Caution with other serotonergic drugs such as amphetamine derivatives, dextromethorphan, meperidine, St. John's wort, tryptophan, etc., due to risk of serotonin syndrome. Increased risk of GI bleeding with NSAIDs or antiplatelet agents. Consider other risk	Always take with food to maximize efficacy.

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
				factors for GI bleeding.	
Drug Class: Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs)					
duloxetine Cymbalta, generics < $40–80	GAD	Initial: 30 mg/day PO Target: 60–120 mg/day PO	CNS: anxiety, agitation, insomnia, headache, extrapyramidal effects. GI: nausea, vomiting, diarrhea, constipation, increased risk of upper GI bleeding. Cardiovascular: hypertension, tachycardia, orthostatic hypotension (especially in elderly). Others: dry mouth, increased sweating, sexual dysfunction, SIADH with hyponatremia.	Metabolized by CYP1A2 and CYP2D6. Caution with drugs that may alter its metabolism. It is also a moderate CYP2D6 inhibitor. Combination with MAOI is contraindicated due to an increased risk of serotonin syndrome. Caution with other serotonergic drugs such as amphetamine derivatives, dextromethorphan, meperidine, St. John's wort and tryptophan due to risk of serotonin syndrome. Increased risk of GI bleeding with the combination of an SNRI and an NSAID or an antiplatelet agent. Consider other risk factors for GI bleeding.	Monitoring blood pressure and heart rate once a week is recommended during first month of treatment and when increasing doses. Contraindicated if ClCr <30 mL/min.
venlafaxine extended-release Effexor XR, generics < $40	AG, PD, GAD, SAD	Initial: 37.5 mg/day PO Target: 75–300 mg/day PO	CNS: anxiety, agitation, insomnia, headache, extrapyramidal effects. GI: nausea, vomiting, diarrhea, constipation, increased risk of upper GI bleeding. Cardiovascular: hypertension, tachycardia, orthostatic hypotension (especially in elderly). Others: dry mouth, increased sweating, sexual dysfunction, SIADH with hyponatremia.	Metabolized by CYP2D6 and CYP3A4. Caution with drugs that may alter its metabolism. It is also a weak CYP2D6 inhibitor. Concurrent use with MAOI is contraindicated due to an increased risk of serotonin syndrome. Caution with other serotonergic drugs such as amphetamine derivatives, dextromethorphan, meperidine, St. John's wort and tryptophan due to risk of serotonin syndrome. Increased risk of GI bleeding with the combination of an SNRI and an NSAID or an antiplatelet agent. Consider other risk factors for GI bleeding.	Monitoring blood pressure and heart rate once a week is recommended during first month of treatment and when increasing doses. A dose higher than 150 mg/day is required to influence norepinephrine.
Drug Class: Tricyclic Antidepressants (TCAs)					
clomipramine Anafranil, generics	AG, PD	Initial: 10–25 mg/day PO	CNS: drowsiness, psychomotor impairment, agitation (especially on	TCA are all metabolized by the liver. Caution with drugs that may	TCAs are toxic in overdose; contraindicated in

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
$40–120		Target: 100–250 mg/day PO	initiation of therapy), headache, myoclonus, seizure. Anticholinergic: dry mouth, constipation, blurred vision, urinary retention, cognitive impairment. Cardiovascular: orthostatic hypotension, palpitations, dizziness, tachycardia, arrhythmias, QT_c interval prolongation. Other: increased appetite, weight gain, sexual dysfunction.	alter their metabolism. They are also CYP2D6 inhibitors; imipramine also inhibits CYP2C19. Do not use in combination with MAOI due to an increased risk of serotonin syndrome. Increased risk of additive adverse effects when combined with other anticholinergic drugs or CNS depressants such as alcohol. Use with caution with other drugs associated with prolonged QT_c interval or other risk factors. Caution with drugs that may lower the seizure threshold with concurrent use of a TCA.	case of suicidality. Use with caution in case of cardiovascular disease, cognitive impairment or history of seizure. Consider risk factors for QT_c prolongation when using a TCA.
desipramine generics < $40–80	AG, PD	Initial: 25 mg/day PO Target: 100–300 mg/day PO	CNS: drowsiness, psychomotor impairment, agitation (especially on initiation of therapy), headache, myoclonus, seizure. Anticholinergic: dry mouth, constipation, blurred vision, urinary retention, cognitive impairment. Cardiovascular: orthostatic hypotension, palpitations, dizziness, tachycardia, arrhythmias, QT_c interval prolongation. Other: increased appetite, weight gain, sexual dysfunction.	TCA are all metabolized by the liver. Caution with drugs that may alter their metabolism. They are also CYP2D6 inhibitors; imipramine also inhibits CYP2C19. Do not use in combination with MAOI due to an increased risk of serotonin syndrome. Increased risk of additive adverse effects when combined with other anticholinergic drugs or CNS depressants such as alcohol. Use with caution with other drugs associated with prolonged QT_c interval or other risk factors. Caution with drugs that may lower the seizure threshold with concurrent use of a TCA.	TCAs are toxic in overdose; contraindicated in case of suicidality. Use with caution in case of cardiovascular disease, cognitive impairment or history of seizure. Consider risk factors for QT_c prolongation when using a TCA.
imipramine generics < $40–80	AG, GAD, PD	Initial: 10–25 mg/day PO Target: 100–300 mg/day PO	CNS: drowsiness, psychomotor impairment, agitation (especially on initiation of therapy), headache, myoclonus, seizure. Anticholinergic: dry mouth, constipation, blurred vision, urinary retention,	TCA are all metabolized by the liver. Caution with drugs that may alter their metabolism. They are also CYP2D6 inhibitors; imipramine also inhibits CYP2C19. Do not use in combination with	TCAs are toxic in overdose; contraindicated in case of suicidality. Use with caution in case of cardiovascular disease, cognitive impairment or history of seizure.

Drug/Cost[a]	Indications[b]	Dosage	Adverse Effects	Drug Interactions	Comments
			cognitive impairment. Cardiovascular: orthostatic hypotension, palpitations, dizziness, tachycardia, arrhythmias, QT_c interval prolongation. Other: increased appetite, weight gain, sexual dysfunction.	MAOI due to an increased risk of serotonin syndrome. Increased risk of additive adverse effects when combined with other anticholinergic drugs or CNS depressants such as alcohol. Use with caution with other drugs associated with prolonged QT_c interval or other risk factors. Caution with drugs that may lower the seizure threshold with concurrent use of a TCA.	Consider risk factors for QT_c prolongation when using a TCA.

[a] Cost of 30-day supply at target dose; includes drug cost only.
[b] Indications supported by clinical evidence; not all Health Canada-approved indications.
Dosage adjustment may be required in renal impairment; see Dosage Adjustment in Renal Impairment.
Abbreviations: AG = agoraphobia; CNS = central nervous system; CYP = cytochrome P450; GAD = generalized anxiety disorder; GI = gastrointestinal; NSAID = nonsteroidal anti-inflammatory drug; PD = panic disorder; SAD = social anxiety disorder (social phobia); SIADH = syndrome of inappropriate antidiuretic hormone

Suggested Readings

Blanco C, Bragdon LB, Schneier FR et al. The evidence-based pharmacotherapy of social anxiety disorder. *Int J Neuropsychopharmacol* 2013;16(1):235-49.

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